Exercise 8.4 SOLUTION

1. using binomial theorem find the value of the following to three places of decimals:

i) $\sqrt{99}$

Sol: $\sqrt{99} = (99)^{\frac{1}{2}} = (100-1)^{\frac{1}{2}} = 100^{\frac{1}{2}}\left(1-\frac{1}{100}\right)^{\frac{1}{2}}$

$$= 10\left[1+\frac{1}{2}(-0.01)+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}(-0.01)^2+....\right]$$

$$= 10\left[1+0.005+\frac{0.5(-0.5)}{2}(-0.0001)+....\right]$$

$$= 10[1+0.005+0.000125+....] = 10(0.9499) = 9.499\,approx.$$

نوٹ

دیے گئے جذر یا Power کے اندر نمبر سے اس طرح دو نمبر بنائیں کہ پہلا نمبر بڑا ہو اور اُسے بغیر کیلکولیٹر سے آپ حل کر سکیں۔ جب آپ اسے common لیں۔

ii) $(1.03)^{\frac{1}{3}}$

Sol: $(1.03)^{\frac{1}{3}} = (1+0.03)^{\frac{1}{3}} = 1+\frac{1}{3}(0.03)+\frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}(0.03)^2+....$

$$= 1+0.01+\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)\left(\frac{-2}{3}\right)(0.0009)+....$$

$$= 1+0.01-0.0001+.... = 1.010(approx)$$

iii) $\frac{1}{\sqrt[5]{252}}$

Sol: $\frac{1}{\sqrt[5]{252}} = \frac{1}{(252)^{1/5}} = (252)^{-1/5} = (243+9)^{-1/5}$

$$= (243)^{-1/5}\left(1+\frac{9}{243}\right)^{-1/5} = \frac{1}{(243)^{1/5}}(1+0.037)^{-1/5}$$

$$= \frac{1}{3^{5\times\frac{1}{5}}}\left[1+\left(-\frac{1}{5}\right)(0.037)+\frac{-\frac{1}{5}\left(-\frac{1}{5}-1\right)}{2!}(0.037)^2+....\right]$$

$$= \frac{1}{3}(1-0.0074+0.000167+....) = 0.331(approx)$$

iv) $\frac{\sqrt{7}}{\sqrt{8}}$

Sol: $\frac{\sqrt{7}}{\sqrt{8}} = \left(\frac{7}{8}\right)^{\frac{1}{2}} = \left(1-\frac{1}{8}\right)^{\frac{1}{2}}+.... = 1+\frac{1}{2}\left(\frac{-1}{8}\right)-\frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)}{2!}\left(\frac{-1}{8}\right)^2+....$

$$= 1-0.0625+\frac{1}{2}\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(\frac{1}{64}\right)+....$$

$$= 1-0.0625-0.001953 = 0.935(approx)$$

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2. **Find the remainder when 8^{100} is divided by 7.**

Sol:

$$8^{100} = (7+1)^{100}$$

$$= \binom{100}{0}7^{100}\ 1^0 + \binom{100}{1}7^{99}\ 1^1 + \binom{100}{2}7^{98}\ 1^2 + \ldots\ldots + \binom{100}{99}7^1\ 1^{99} + \binom{100}{100}7^0\ 1^{100}$$

$$= 7^{100} + \binom{100}{1}7^{99} + \binom{100}{2}7^{98} + \ldots. + \binom{100}{99}7 + 1 = 7\left(7^{99} + \binom{100}{1}7^{98} + \binom{100}{2}7^{97} + \ldots. + \binom{100}{99}\right) + 1$$

$= 7k + 1$ (Where k is an integer resulting from the terms with a factor of 7)

Hence remainder is 1.

2nd Method

$$8 \equiv 1 \pmod 7$$

$$8^{100} \equiv 1^{100} \pmod 7$$

$$8^{100} \equiv 1 \pmod 7$$

Remainder is 1.

3. **Find the remainder when 2^{100} is divided by 3.**

Sol: $2^{100} = (3-1)^{100}$

$$= \binom{100}{0}3^{100}(-1)^0 + \binom{100}{1}3^{99}(-1)^1 + \binom{100}{2}3^{98}(-1)^2 + \ldots + \binom{100}{99}3^1(-1)^{99} + \binom{100}{100}3^0(-1)^{100}$$

$$= 3^{100} - \binom{100}{1}3^{99} + \binom{100}{2}3^{98} + \ldots. - \binom{100}{99}3 + 1$$

$$= 3\left(3^{99} - \binom{100}{1}3^{98} + \binom{100}{2}3^{97} + \ldots. - \binom{100}{99}\right) + 1$$

$= 3m + 1$ (where m is an integer resulting from the terms with factor 3)

Hence remainder is 1.

2nd Method

Sol: $2 \equiv -1 \pmod 3$

$$2^{100} \equiv -1 \pmod 3$$

Remainder is 1

4. **Using the binomial theorem, find which number is larger:**

i) $\mathbf{19^{10} + 20^{10}}$ ***or*** $\mathbf{21^{10}}$ $\mathbf{29^{15} + 30^{15}}$ ***or*** $\mathbf{31^{15}}$

Sol:

i) $\mathbf{19^{10} + 20^{10}}$ ***or*** $\mathbf{21^{10}}$

We want to compare 20^{10} *and* $21^{10} - 19^{10}$

Now, $21^{10} - 19^{10} = (20+1)^{10} - (20-1)^{10}$

$$= \binom{10}{0}20^{10}\ 1^0 + \binom{10}{1}20^9\ 1^1 + \binom{10}{2}20^8\ 1^2 + \ldots. + \binom{10}{9}20^1\ 1^9 + \binom{10}{10}20^0\ 1^{10}$$

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$$-\left[\binom{10}{0}20^{10}(-1)^0+\binom{10}{1}20^9(-1)^1+\binom{10}{2}20^8(-1)^2+....+\binom{10}{9}20^1(-1)^9+\binom{10}{10}20^0(-1)^{10}\right]$$

$$=\cancel{20^{10}}+\binom{10}{1}20^9+\cancel{\binom{10}{2}20^8}+...+\binom{10}{9}20+\cancel{1}-\cancel{20^{10}}+\binom{10}{1}20^9-\cancel{\binom{10}{2}20^8}+\binom{10}{3}20^7+...+\binom{10}{9}20-\cancel{1}$$

$$=2\left[\binom{10}{1}20^9+\binom{10}{3}20^7+\binom{10}{5}20^5+\binom{10}{7}20^3+\binom{10}{9}20\right]$$

$$(20)20^9+(240)20^7+(504)20^5+(240)20^3+(20)20=20^{10}+positive\ term$$

$$\Rightarrow 21^{10}-19^{10}>20^{10}\ or\ 21^{10}>19^{10}+20^{10}$$

Hence 21^{10} is larger.

ii) **$29^{15}+30^{15}$ or 31^{15}**

Sol: $31^{15}-29^{15}=(30+1)^{15}-(30-1)^{15}$

$$\left[\binom{15}{0}30^{15}.1^0+\binom{15}{1}30^{14}.1^1+\binom{15}{2}30^{13}.1^2+\binom{15}{3}30^{12}.1^3+...\right]$$

$$-\left[\binom{15}{0}30^{15}.(-1)^0+\binom{15}{1}30^{14}.(-1)^1+\binom{15}{2}30^{13}.(-1)^2+\binom{15}{3}30^{12}.(-1)^3+...\right]$$

$$=\left[30^{15}+15.30^{14}+\binom{15}{2}30^{13}+\binom{15}{3}30^{12}+...\right]-\left[30^{15}-15.30^{14}+\binom{15}{2}30^{13}-\binom{15}{3}30^{12}+...\right]$$

$$=2\left(15.30^{14}\right)+2\binom{15}{3}13^{12}+...$$

$=30.30^{14}+k$, where k is some positive constant

$=30^{15}+k$

$31^{15}-29^{15}>30^{15}$

$31^{15}>29^{15}+30^{15}$

Hence 31^{15} is greater then $29^{15}+30^{15}$

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5. Using the binomial theorem, show that:

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i) **5^7+7^5 is divisible by 36**

Sol: $5^7=(6-1)^7$

$$=\binom{7}{0}6^7(-1)^0+\binom{7}{1}6^6(-1)^1+\binom{7}{2}6^5(-1)^2+...+\binom{7}{6}6^1(-1)^6+\binom{7}{7}6^0(-1)^7$$

$$6^2\left[6^5+\binom{7}{1}6^4(-1)^1+\binom{7}{2}6^3(-1)^2+\binom{7}{3}6^2(-1)^3+\binom{7}{4}6(-1)^4+\binom{7}{5}(-1)^5\right]7\times6-1$$

$$=36k_1+42-1=36k_1+41$$

$$=36k_1+36+5=36(k_1+1)+5\qquad(i)$$

Again $7^5=(6+1)^5=\binom{5}{0}6^51^0+\binom{5}{1}6^41^1+\binom{5}{2}6^31^2+...+\binom{5}{4}6^11^6+\binom{5}{7}6^01^7$

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$$6^2\left[6^3+\binom{5}{1}6^2+\binom{5}{2}6^1+\binom{5}{3}\right]+5\times6+1=36k_2+31$$

$$=36k_2+36-5=36(k_2+1)-5 \qquad (ii)$$

Now (i) + (ii)

$$7^5+5^7=36(k_1+1)+\cancel{5}+36(k_2+1)-\cancel{5}$$

$$=36(k_1+1+k_2+1)$$

Hence 7^5+5^7 *is divisible by* 36

i) **$(17)^{15}+(13)^7$ is divisible by 6**

Sol: $(17)^{15}=(18-1)^{15}$

$$=\binom{15}{0}18^{15}(-1)^0+\binom{15}{1}18^{14}(-1)^1+\binom{15}{2}18^{13}(-1)^2+\ldots+\binom{15}{14}18^1(-1)^{14}+\binom{15}{15}18^0(-1)^{15}$$

$$=18^{15}-15(18)^{14}+\binom{15}{2}18^{13}+\ldots+\binom{15}{14}18-1$$

Similarly, $(13)^7=(12+1)^7$

$$=\binom{7}{0}12^7+\binom{7}{1}12^6+\binom{7}{2}12^5+\ldots+\binom{7}{6}12^1+\binom{7}{7}12^0$$

$$=12^7+7(12)^6+\binom{7}{2}12^5+\ldots+\binom{7}{6}12+1$$

By Adding we get

$$(17)^{15}+(13)^7=18\left\{(18)^{14}-15(18)^{13}+\binom{15}{2}(18)^{12}+\ldots\binom{15}{14}\right\}-\cancel{1}+12\left\{(12)^6+7(12)^5+\binom{7}{2}(12)^4+\ldots+\binom{7}{6}\right\}+\cancel{1}$$

$$=18k_1+12k_2, \text{where } k_1,k_2\in z$$

$$=6(3k_1+2k_2)$$

Hence $(17)^{15}+(13)^7$ is divisible by 6

iii) **$(21)^9+(19)^{11}$ is divisible by 20**

Sol: $(21)^9=(20+1)^9$

$$=\binom{9}{0}.20^9+(9)20^8+\binom{9}{2}20^7+\ldots+\binom{9}{8}20^1+\binom{9}{9}20^0$$

$$20^9+9.20^8+\binom{9}{2}20^7+\ldots+9.20+1$$

Similarly, $(19)^{11}=(20-1)^{11}$

$$=\binom{11}{0}.20^{11}(-1)^0+\binom{11}{1}20^{10}(-1)^1+\binom{11}{2}20^9(-1)^2+\ldots+\binom{11}{10}20^1(-1)^{10}+\binom{11}{11}20^0(-1)^{11}$$

$$=\binom{11}{0}.20^{11}-\binom{11}{1}20^{10}+\binom{11}{2}20^9+\ldots+\binom{11}{10}20^1-\binom{11}{11}20^0$$

$$20^{11}-11.20^{10}+\binom{11}{2}20^9+\ldots+11.20-1$$

By adding we get

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$(21)^9+(21)^{11}=20\{(20^8+9.20^7+36.20^6+...+9)-1+(20^{10}-11.20^9+55.20^8+...+11)-1\}$

$(21)^9+(21)^{11}=20k$, where $k\in z$

Hence $(21)^9+(19)^{11}$ is divisible by 20.

iv) **$(31)^4+(29)^6$ is divisible by 30**

Sol: $(31)^4=(30+1)^4$

$=\binom{4}{0}.30^4+\binom{4}{1}30^3+\binom{4}{2}30^2+\binom{4}{3}30^1+\binom{4}{4}30^0$

$30(30^3+4.30^2+6.30+4)+1$

Similarly, $(29)^6=(30-1)^6$

$=\binom{6}{0}30^6(-1)^0+\binom{6}{1}30^5(-1)^1+\binom{6}{2}30^4(-1)^2+...+\binom{6}{5}30^1(-1)^5+\binom{6}{6}30^0(-1)^6$

$=\binom{6}{0}30^6-\binom{6}{1}30^5+\binom{6}{2}30^4+...-\binom{6}{5}30^1+\binom{6}{6}30^0$

$30(30^5-6.30^4+15.30^3...-6)-1$

By adding we get

$(31)^4+(29)^6=30\{(30^3+4.30^2+6.30+4)+(30^5-6.30^4+15.30^3...+6)\}$

$30k$, where $k\in z$

Hence $(31)^4+(29)^6$ is divisible by 30.

v) **$(101)^5+(99)^7$ is divisible by 100**

Sol: $(101)^5=(100+1)^5$

$=\binom{5}{0}100^5+\binom{5}{1}100^4+\binom{5}{2}100^3+...+\binom{5}{4}100^1+\binom{5}{5}100^0$

$100^5+5.100^4+10.100^3+...+5.100+1$

$100(100^4+5.100^3+10.100^2+...+5)+1$

Similarly, $(99)^7=(100-1)^7$

$=\binom{7}{0}100^7(-1)^0+\binom{7}{1}100^6(-1)^1+\binom{7}{2}100^5(-1)^2+...+\binom{7}{6}100^1(-1)^6+\binom{7}{7}100^0(-1)^7$

$=\binom{7}{0}100^7-\binom{7}{1}100^6+\binom{7}{2}100^5-...+\binom{7}{6}100^1-\binom{7}{7}100^0$

$100(100^6-7.100^5+21.100^4-...+7)-1$

By adding we get

$(101)^5+(99)^7=100\{(100^4+5.100^3+10.100^2+...+5)-1+(100^6-7.100^5+21.100^4-...+7)-1\}$

$=100k$, where $k\in z$

Hence $(101)^5+(99)^7$ is divisible by 100.

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6. Using the binomial theorem, find the remainder when 3^{101} is divided by 8.

Sol: $3^{101} = 3^{100}.3^1 = \left(3^2\right)^{50}.3$

$= 9^{50}.3 = (8+1)^{50}.3$

$= \left[\binom{50}{0}8^{50}1^0 + \binom{50}{1}8^{49}1^1 + \binom{50}{2}8^{48}1^2 + \ldots + \binom{50}{49}8^1\ 1^{49} + \binom{50}{50}8^0 1^{50}\right].3$

$= \left[8^{50} + \binom{50}{1}8^{49} + \binom{50}{2}8^{48} + \ldots + \binom{50}{49}8 + 1\right].3$

$= (8k+1).3$

$24k + 3 = 8(3k) + 3$

$\Rightarrow 3^{101} = 8(3k) + 3$

When 3^{101} is divided by 8 then remainder is 3

2nd Method

Sol: $3^{101} = 3.\left(3^2\right)^{50} = 3.9^{50} = 3.(1+8)^{50}$

$= 3\sum_{k=0}^{50}\binom{50}{k}1^{50-k}8^k = 3\left(\binom{50}{0}8^0 + \binom{50}{1}8^1 + \binom{50}{2}8^2 + \ldots\right) = 3\left(1 + 50.8 + 64.\binom{50}{2} + \ldots\right)$

$3^{101} \equiv 3(1+0+0+\ldots)(\text{mod}\,8)$

$3^{101} \equiv 3(\text{mod}\,8)$

7. Using the binomial theorem, find the last digit of the number $(32)^{32}$.

Sol: $(32)^{32} = (300+2)^{32}$

$= \binom{32}{0}300^{32}.2^0 + \binom{32}{1}30^{31}.2^1 + \ldots + \binom{32}{31}30^1.2^{31} + \binom{32}{32}30^0.2^{32}$

$1.30^{32}.(1) + 32.30^{31}.2 + \ldots + 31(30).2^{31} + (1)(1).2^{32}$

$\left(30^{32} + 64.30^{31} + \ldots + 32.30.2^{31}\right) + 2^{32}$

$10k_1 + 2^{32}$, where $k_1 \in z$

$\Rightarrow$ last digit in $(2)^{32}$

Now, $2^{32} = 2^2.2^{30} = 4.\left(2^5\right)6 = 4(32)^6 = 4(30+2)^6$

$= 4\left\{\binom{6}{0}30^6.2^0 + \binom{6}{1}30^5.2^1 + \ldots + \binom{5}{5}30^1.2^5 + \binom{6}{6}30^0.2^6\right\}$

$= 4\left\{\left(30^6 + 6.30^5.2 + \ldots + 6.30.2^5\right) + 2^6\right\} = 4\left(10k_2 + 2^6\right)$, where $k_2 \in z$

$= 10k_2 + 4.2^6$

$\Rightarrow$ last digit in $(2)^{32}$ = last digit in 4.2^6

= last digit in 256

= 6

Therefore, last digit in $(32)^{32} = 6$.

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8. **Using the binomial theorem, show that $7^n - 6n$ leaves remainder 1 when divided by 7 for all positive integers n**

Sol: $7^n = (1+6)^n = \sum_{k=0}^{n}\binom{n}{k}1^{n-k}6^k = \binom{n}{0}6^0 + \binom{n}{1}6^1 + \binom{n}{k}6^k$

$$7^n = 1 + 6n + 6^2\binom{n}{2} + 6^3\binom{n}{3} + \ldots$$

$$7^n - 6n = 1 + 6^2\binom{n}{2} + 6^3\binom{n}{3} + \ldots$$

$$7^n - 6n = 1 + 6\left(6\binom{n}{2} + 6^2\binom{n}{3} + \ldots\right)$$

$$7^n - 6n \equiv 1 \pmod 6$$

9. **By using binomial theorem show that for each $n \in N, 5^n - 1$ is divisible by 4.**

Sol: we want to show $4 \mid (5^n - 1)$ for $n \in N$

Using $(4+1)^n = \sum_{k=0}^{n}\binom{n}{k}4^{n-k}1^k$:

$$5^n = 4^n + n.4^{n-1} + \ldots + 4n + 1$$

$$5^n - 1 = 4^n + n.4^{n-1} + \ldots + 4n$$

$$5^n - 1 = 4\left(4^n + n.4^{n-1} + \ldots + n\right)$$

Since the term in the parenthesis is an integer $4 \mid (5^n - 1)$.

10. **By using binomial theorem show that for each $n \in N, 5^n - 2^n$ is divisible by 3.**

Sol: $5^n = (2+3)^n$

$$5^n = \binom{n}{0}2^n.3^0 + \binom{n}{1}2^{n-1}.3^1 + \binom{n}{2}.2^{n-2}.3^2 + \binom{n}{3}.2^{n-3}.3^3 + \ldots + \binom{n}{n}2^0.3^n$$

$$5^n = \binom{n}{0}2^n + n.2^{n-1}.3 + \binom{n}{2}.2^{n-2}.3^2 + \binom{n}{2}2^{n-3}.3^3 + \ldots + 3^n$$

$$5^n - 2^n = 3n.2^{n-1} + \binom{n}{2}.2^{n-2}.3^2 + \binom{n}{2}2^{n-3}.3^3 + \ldots + 3^n$$

$$= 3\left[n.2^{n-1} + \binom{n}{2}.2^{n-2}.3 + \binom{n}{2}.2^{n-3}.3^2 + \ldots + 3^{n-1}\right]$$

$= 3k$, Where $k \in z$

= Multiplies of 3

Hence, $5^n - 2^n$ is divisible by 3 for all $n \in N$.

11. **Show that $a^2 + (a+2)^2 + (a+4)^2 + 1$ is divisible by 12, whenever "a" is an odd integer.**

Sol: $E = a^2 + (a+2)^2 + (a+4)^2 + 1$

$$E = a^2 + (a^2 + 4a + 4) + (a^2 + 8a + 16) + 1$$

$$E = 3a^2 + 12a + 21$$

Given a is an odd integer, let $a = 2k+1$ for some integer k.

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$$E = 3(2k+1)^2 + 12(2k+1) + 21$$

$$E = 3(4k^2 + 4k + 1) + 24k + 12 + 21$$

$$E = 12k^2 + 12k + 3 + 24k + 33$$

$$E = 12k^2 + 36k + 36$$

$$E = 12(k^2 + 3k + 3)$$

Since k is an integer, $k^2 + 3k + 3$ is an integer.
Therefore, E is divisible by 12

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12. **A company expects its manual revenue to grow at a fixed rate of 6% per year. The revenue in year 1 is R=Rs. 10,000,000. Estimate the company's revenue after 4 years using the binomial theorem for small growth rates.**

Sol: **Company Revenue Growth**

$$R_n = R_0(1+r)^n$$

$$R_0 = 10{,}000{,}000$$

$$r = 0.06$$

$$n = 4$$

$$R_4 = 10{,}000{,}000(1+0.06)^4$$

$$(1+0.06)^4 \approx 1 + 4(0.06) + \frac{4.3}{2}(0.06)^2 + \frac{4.3.2}{6}(0.06)^3$$

$$(1+0.06)^4 \approx 1 + 0.24 + 6(0.0036) + 4(0.000216)$$

$$(1+0.06)^4 \approx 1 + 0.24 + 0.0216 + 0.000864$$

$$(1+0.06)^4 \approx 1.262464$$

$$R_4 = 10{,}000{,}000 \times 1.262464$$

$$R_4 = 12{,}624{,}640$$

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13. **A bank offers a compound interest rate of 10% per year. Zafar invests Rs. 2,000,000 for 4 years. How much will his investment be worth at the end of 4 years?**

Sol: **Zafar's compound interest investment**

$$A = P(1+r)^n$$

$$P = 2{,}000{,}000$$

$$r = 0.10$$

$$n = 4$$

$$A = 2{,}000{,}000(1+0.10)^4$$

$$A = 2{,}000{,}000(1.10)^4$$

$$(1.10)^4 = 1.4641$$

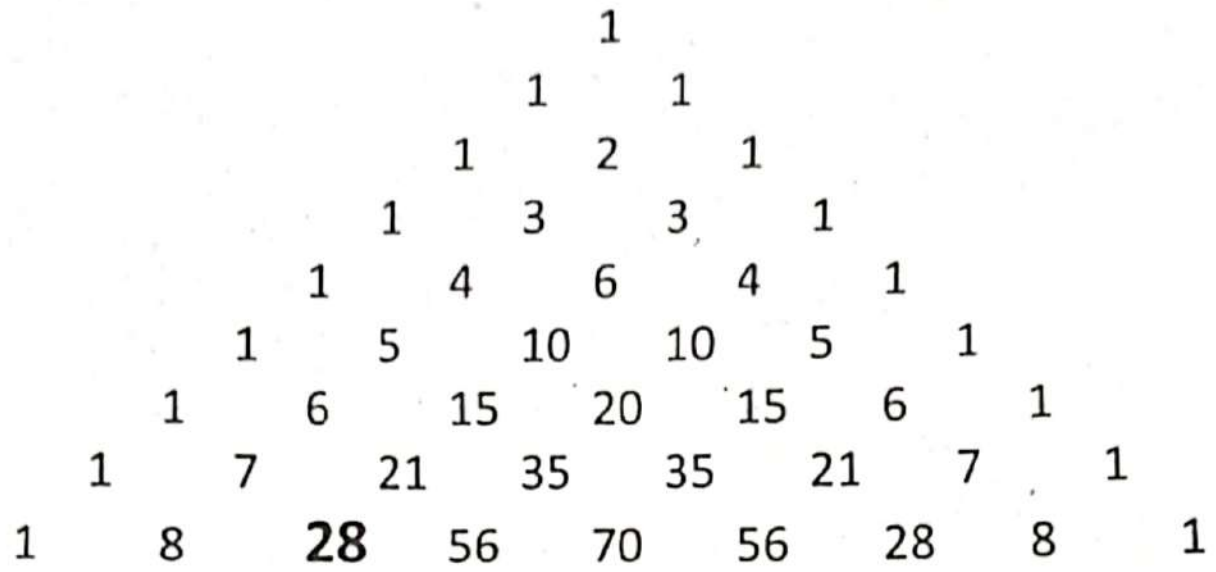
$$A = 2{,}000{,}000 \times 1.4641$$

$$A = 2{,}928{,}200$$

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14. **Zaid is organizing a sports competition with 8 teams. Every team plays against every other team exactly once. How many matches will be played in total? Use Pascal's triangle to solve this.**

Sol: To find the number of matches, locate the 8th row of Pascal's triangle and the 3rd number in that row (Corresponding to choosing 2 teams).



The number of matches is **28**.